Université d'Ottawa · University of Ottawa
Faculté des sciences Mathématiques et de statistique
Faculty of Science Mathematics and Statistics

Introduction to Linear Algebra MAT 1341C

Final Exam

April 13, 2025, 14:00-17:00

You must **sign below** to confirm that you have read, understand, and will follow these **instructions**:

- This is a 3-hour **closed-book** exam; no notes are allowed. **Calculators are not permitted.**
- The exam consists of 8 questions on 15 pages. Page 15 provides additional work space. If you need additional space, you can use the backs of any of the pages. Indicate clearly where your answer can be found if it is not written down immediately after where the question is given.
- Questions 1–3 are short-answer questions. No justification is required for these questions.
- Questions 4–8 are long-answer questions. You must show all relevant steps and give appropriate justifications in order to obtain full marks.
- On Page 14, there is a bonus question. Only attempt it after you have finished all the other questions.
- Cellular phones, smart watches , unauthorized electronic devices, or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

Last name:_____

First name:_____

Student Number:_____

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Q. 1	Q. 2	Q. 3	Q. 4	Q. 5	Q. 6	Q. 7	Q. 8	Bonus	Total
/16	/6	/10	/12	/12	/16	/16	/12	/5	/100

(1) Determine whether each of the following statements is true or false.

(a) The polynomial $p(x) = 1341x^{1341} - 1741x^{20} + 5$ has at least one root in $\mathbb{C}$. *Solution:* T

(b) If a vector space V contains a linearly independent set $\{v_1, \ldots, v_{10}\}$, then $\dim(V) \leq 10$. *Solution:* F

(c) If V is a vector space, W is a subspace of V, and $v_1, v_2, v_3 \in W$, then $v \in W$ for every $v \in \mathrm{Span}\{v_1, v_2, v_3\}$. *Solution:* T

(d) If U is a subspace of $\mathbb{R}^3$ such that $\dim(U^\perp) = 2$, then U is a plane. *Solution:* F

(e) A linear system $Ax = b$, where $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$, has a solution if and only if $\mathrm{rank}([A|b]) > \mathrm{rank}(A)$. *Solution:* F

(f) If $A, B \in M_n(\mathbb{R})$ are invertible, then $AB = BA$. *Solution:* F

(g) If $A \in M_n(\mathbb{R})$ with $\det(A) = 0$, then $A^n = \mathbf{0}$. *Solution:* F

(h) If a matrix is diagonalizable, then it is invertible. *Solution:* F

(i) Let U be a subspace of $\mathbb{R}^n$. If $\mathbf{w}$ is a vector in $\mathbb{R}^n$ that does not belong to U, then $\|\mathrm{Proj}_U(\mathbf{w})\| < \|\mathbf{w}\|$. *Solution:* T

(j) If $A \in M_n(\mathbb{R})$ has n distinct eigenvalues, then A is diagonalizable. *Solution:* T

Below, circle T if the statement is true or F if the statement is false. [20]

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
T/F	T/F	T/F	T/F	T/F	T/F	T/F	T/F	T/F	T/F

(2) (a) Among the following vector spaces, which have dimension 2? Indicate ***all*** correct answers.

A. $\{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$.

B. $\{f \in \mathbb{P}_3 : f(1) = f(2) = 0\}$, where $\mathbb{P}_3$ denotes the set of polynomials of degree at most 3.

C. $\{A \in M_2 : A = A^T\}$.

D. $\mathrm{Span}\{(1, 1, 1), (1, 2, 3), (0, 1, 2)\}$.

E. The column space of $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

Answers: [ ] [5]

Solution: A,B,D

(b) Let $A = \begin{bmatrix} 1 & 5 & 2 \\ 3 & 7 & 1 \\ 2 & 5 & 1 \end{bmatrix}$, an invertible matrix. Among the following statements, which ones are true? Indicate ***all*** correct answers.

A. The second row of the matrix A^{-1} is $\begin{bmatrix} 1 & 3 & -7 \end{bmatrix}$.

B. The second row of the matrix A^{-1} is $\begin{bmatrix} 1 & 3 & -5 \end{bmatrix}$.

C. The third row of the matrix A^{-1} is $\begin{bmatrix} -1 & -5 & 8 \end{bmatrix}$.

D. The third row of the matrix A^{-1} is $\begin{bmatrix} -1 & 5 & -8 \end{bmatrix}$.

E. The three rows of A form a basis of $\mathbb{R}^3$.

Answers: [ ] [5]

Solution: B,C,E

(3) Let A be a matrix of size 12×12 and B be a matrix of size 12×9. For each statement below, write the value of b (or the values of b and d) such that the statement is true.

(a) Null(B^T) is a subspace of $\mathbb{R}^b$. b=_____ [2]

Solution: 12

(b) If rank$(B) = 8$, then the dimension of Null(B^T) is b. b=_____ [2]

Solution: 4

(c) AB is a matrix of size $b \times d$. b=_____, d=_____ [2]

Solution: 12, 9

(d) If $\det(A) = 2$, then $\det(A^{-1}) = b$ and $\det(A^T) = d$. b=_____, d=_____ [2]

Solution: 1/2,2

(e) $\dfrac{4-6i}{1+i} = b + di$. b=_____, d=_____ [2]

Solution: -1, -5

(4) Consider the following linear system, where a and b are parameters:

$$\begin{aligned} x + y + z &= 1 \\ x + 2y - z &= b \\ 5x + 7y + az &= b \end{aligned}$$

(a) For which values of a and b does the system:

(i) have a unique solution?

(ii) have no solution?

(iii) have infinitely many solutions? [6]

Solution:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & b \\ 5 & 7 & a & b \end{bmatrix} \quad \begin{matrix} R_2 - R_1 \to \\ R_3 - 5R_1 \to \end{matrix} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & b-1 \\ 0 & 2 & a-5 & b-1 \end{bmatrix} \quad R_3 - 2R_2 \to \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & b-1 \\ 0 & 0 & a-1 & 1-b \end{bmatrix}$$

(i) Unique solution: $a \neq 1$

(ii) No solution: $a = 1$

(iii) Infinitely many solutions: $a = 1$ and $b = 1$.

(b) For which values of a and b is the augmented matrix of the system of rank 3? [2]
Solution: Using the row-equivalent matrix from (a), we see that the augmented matrix has rank 3 iff $a \neq 1$ or $b \neq 1$.

(c) Let A be the 3×3 coefficient matrix of the given linear system. Find a basis of the null space (kernel) ***and*** determine its dimension when $a = 1$. [4]

Solution: We set $a = 1$ and find solutions to $Av = \vec{0}$. Using the row-reduction from (a), we have that A is row-equivalent to

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \qquad R_1 - R_2 \to \qquad \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

The solutions are therefore given by $z = t$, $y = 2t$, $x = -3t$, or

$$v = t \begin{bmatrix} -3t \\ 2t \\ t \end{bmatrix} = t \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

A basis for the null space is therefore $\left\{ \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} \right\}$.

Since there is only one vector in this basis, the dimension of null space is 1.

(5) Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ -1 & -1 & 5 & -3 \\ 2 & 4 & x+2 & 8 \\ 1 & 5 & 7-x & x+2 \end{pmatrix}$, where x is a parameter.

(a) Compute the determinant of A. (*Hint: Perform elementary row operations first.*) [7]

Solution:

$$\begin{aligned}
\det(A) &= \begin{vmatrix} 1 & 2 & 3 & 4 \\ -1 & -1 & 5 & -3 \\ 2 & 4 & x+2 & 8 \\ 1 & 5 & 7-x & x+2 \end{vmatrix} \\
&= \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 8 & 1 \\ 0 & 0 & x-4 & 0 \\ 0 & 3 & 4-x & x-2 \end{vmatrix} \begin{matrix} \\ \leftarrow R_2 + R_1 \\ \leftarrow R_3 - 2R_1 \\ \leftarrow R_4 - R_1 \end{matrix} \\
&= 1 \cdot \begin{vmatrix} 1 & 8 & 1 \\ 0 & x-4 & 0 \\ 3 & 4-x & x-2 \end{vmatrix} \\
&= (x-4) \begin{vmatrix} 1 & 1 \\ 3 & x-2 \end{vmatrix} \\
&= (x-4)\big((1)(x-2) - (1)(3)\big) = (x-4)(x-5).
\end{aligned}$$

(In the third line, we evaluate the 4×4 determinant by expanding along the first column. In the next line, we evaluate the 3×3 determinant by expanding along the second row.)

(b) For which values of x is the matrix A invertible? [3]

Solution: A matrix is invertible if and only if $\det(A) \neq 0$. Since we computed

$$\det(A) = (x-4)(x-5),$$

A is invertible for all values of x except $x = 4$ and $x = 5$.

(6) Let $V \subset \mathbb{R}^4$ be the subspace spanned by the linearly independent vectors $\mathbf{v}_1 = (1, 2, 0, 1)$, $\mathbf{v}_2 = (1, 6, 1, -1)$, and $\mathbf{v}_3 = (4, -2, -4, 6)$.

(a) Find an orthogonal basis of V. [7]

Solution: We use the Gram–Schmidt algorithm.

$$\mathbf{u}_1 = \mathbf{v}_1 = (1, 2, 0, 1).$$

Then,

$$\begin{aligned}\mathbf{u}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2}\,\mathbf{u}_1 \\ &= (1, 6, 1, -1) - \frac{12}{6}\,(1, 2, 0, 1) \\ &= (-1, 2, 1, -3).\end{aligned}$$

Finally,

$$\begin{aligned}\mathbf{u}_3 &= \mathbf{v}_3 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2}\,\mathbf{u}_1 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2}\,\mathbf{u}_2 \\ &= (4, -2, -4, 6) - \frac{6}{6} \cdot (1, 2, 0, 1) - \frac{-30}{15}\,(-1, 2, 1, -3) \\ &= (1, 0, -2, -1).\end{aligned}$$

Therefore an orthogonal basis for V is

$$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \{(1, 2, 0, 1), (-1, 2, 1, -3), (1, 0, -2, -1)\}.$$

(b) Calculate the orthogonal projection of $\mathbf{w} = (3, 5, 5, -1)$ onto V. [3]

Solution: We apply the projection formula using the orthogonal basis found in part (a):

$$\begin{aligned}\mathrm{Proj}_V(\mathbf{w}) &= \frac{\mathbf{w}\cdot\mathbf{u}_1}{\|\mathbf{u}_1\|^2}\,\mathbf{u}_1 + \frac{\mathbf{w}\cdot\mathbf{u}_2}{\|\mathbf{u}_2\|^2}\,\mathbf{u}_2 + \frac{\mathbf{w}\cdot\mathbf{u}_3}{\|\mathbf{u}_3\|^2}\,\mathbf{u}_3 \\ &= \frac{12}{6}\,(1,2,0,-1) + \frac{15}{15}\,(-1,2,1,-3) + \frac{-6}{6}\,(1,0,-2,-1) \\ &= (0,6,3,0).\end{aligned}$$

(7) Let $A = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$.

(a) Show that $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ is an eigenvector of A and determine the associated eigenvalue. [3]

Solution: Calculate

$$Av = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since this is the same as $0v$ (and $v \neq \vec{0}$), we conclude that v is an eigenvector with eigenvalue 0.

(b) Find the characteristic polynomial of A and show that the eigenvalues of A are 0 and 1. [4]

Solution: The characteristic polynomial is

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 0 & 2 \\ 0 & 1-\lambda & 0 \\ -1 & 0 & -1-\lambda \end{vmatrix} \\ &= (1-\lambda)\begin{vmatrix} 2-\lambda & 2 \\ -1 & -1-\lambda \end{vmatrix} \\ &= (1-\lambda)\big[(2-\lambda)(-1-\lambda) - 2(-1)\big] \\ &= (1-\lambda)(\lambda^2 - \lambda) \\ &= -\lambda(1-\lambda)^2 \end{aligned}$$

(by expanding along the second row).

Since the roots of this polynomial are 0 and 1, these are the eigenvalues.

(c) Find a basis for the eigenspace E_1. [4]

Solution: We have $E_1 = \mathrm{Null}(A - I)$ and

$$A - I = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ -1 & 0 & -2 \end{bmatrix}.$$

The RREF is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore the basic solutions are $w_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $w_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$. These form a basis for E_1.

(d) Determine the algebraic multiplicity ***and*** the geometric multiplicity of the eigenvalue 1. Justify your answer. [2]

Solution: The characteristic polynomial is $-\lambda(\lambda - 1)^2$, the algebraic multiplicity is the multiplicity of the root 1, which is 2. The geometric multiplicity is $\dim(E_1) = 2$ (by (c)).

(e) Show that A is diagonalizable, and find an invertible matrix $P \in M_3(\mathbb{R})$ and a diagonal matrix $D \in M_3(\mathbb{R})$ such that $P^{-1}AP = D$. [3]

Solution: Since the eigenvalue 0 has algebraic multiplicity 1, it must have geometric multiplicity 1, so the eigenvector v from a must form a basis for E_0.

Combining with the basis w_1, w_2 from (c) therefore gives a basis of eigenvectors.

Put these vectors as columns to form the matrix P, and use the corresponding eigenvalues as the diagonal entries of D, to get:

$$P = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

and

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(8) Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be the linear transformation defined by

$$T(x, y, z) = (x - y, z + 2y, x + z - y, -y).$$

(a) Find the standard matrix of T. [2]

Solution: $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$.

(b) Find a basis for the image $\mathrm{Im}(T)$ of T. [4]

Solution: Since $\mathrm{Im}(T) = \mathrm{Col}(A)$, we use the column space algorithm

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \qquad R_3 - R_1 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_2 + 2R_4 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad R_4 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-R_4 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad R_3 - R_2 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

We see every column has a leading one, so the columns of A form a basis of $\mathrm{Im}(T)$.

(c) Does $(1,1,1,1)$ belong to the image of T? [4]

Solution: Test if $Ax = (1,1,1,1)$ has a solution using the augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 \\ 0 & -1 & 0 & 1 & 1 \end{array}\right] \quad R_3 - R_1 \to \quad \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 1 \end{array}\right]$$

$$-R_2 \to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array}\right]$$

$$R_2 + 2R_4 \to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 \end{array}\right] \quad R_4 - R_3 \to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right].$$

The system is inconsistent, which means that $(1,1,1,1)$ is not in the image of T.

(d) What is the dimension of the kernel $\mathrm{Ker}(T)$ of T? [2]

Solution: The Rank–Nullity Theorem says $\dim \mathrm{Ker}(T) + \dim \mathrm{Im}(T) = 3$. From (b), $\dim \mathrm{Im}(T) = 3$. Therefore, $\dim \mathrm{Ker}(T) = 0$.

Bonus: Let $A \in M_2(\mathbb{R})$ such that $A = A^T$. Prove that A is diagonalizable over $\mathbb{R}$. [5]